

Product Proposal

Car & Bike — India's Most Complete Automotive Marketplace

****Prepared For:**** Prospective Client / Investor

****Prepared By:**** Car & Bike Product Team

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****Platform URL:**** http://localhost:3002 *_(Live demo available)_*

EXECUTIVE SUMMARY

****Car & Bike**** is a full-stack, production-ready automotive super-platform built with Next.js 14, MySQL (Prisma ORM), MongoDB, and Cloudinary. Unlike traditional portals like CarDekho or BikeWale that focus only on browsing and price comparisons, Car & Bike is an ****end-to-end transactional ecosystem**** — connecting individual buyers, sellers, dealerships, resellers, and service stations in one unified platform.

> ****This is not just a listing site. It is a complete automotive commerce platform.****

1. PLATFORM PORTALS

The platform is architected with separate, purpose-built portals for each stakeholder. Below is a complete breakdown based on live system testing.

🤖 1.1 Customer Portal

The customer-facing interface with modern dark/light UI, smooth Framer Motion animations, and location-aware personalization.

| Feature | Description |

|---|---|

| ****Pincode-Based Experience**** | On landing, users enter their 6-digit pincode. The platform validates it, detects their city/state, and personalizes all listings, offers & dealers to their exact area |

| ****New Cars**** | Browse latest brand-new car models with ex-showroom, RTO, insurance & on-road price breakdown per variant |

| ****New Bikes**** | Dedicated section for all new two-wheeler brands and models |

| ****Used Cars / Used Bikes**** | Full marketplace for pre-owned vehicles with owner history, RC details, fitness certificates & insurance data |

| ****Rental Cars / Bikes**** | End-to-end rental marketplace with per-day, per-km, driver, and one-way pricing |

| ****Accessories**** | Shop for vehicle accessories categorized by product type |

| ****Spare Parts**** | Spare parts marketplace with part numbers and compatible model listings |

| ****On-Road Price Calculator**** | Calculates exact on-road price by city using ex-showroom + RTO + insurance + optional charges |

| ****AI Car Assistant Chatbot**** | Integrated 24/7 AI chatbot powered by Twilio that answers car queries and guides buyers |

| ****Test Drive Booking**** | Book a showroom visit or home test drive from any listing page |

| ****Saved Listings (Wishlist)**** | Save favourite vehicles to revisit later |

| ****Vehicle Comparison**** | Side-by-side technical comparison of any two listed vehicles using full specs |

| ****My Garage**** | Personal dashboard showing enquiries, test drive requests, submitted listings, and post-purchase reviews |

| ****Service Station Booking**** | Find nearby authorised service centres and book at-home or at-centre services |

| ****News Section**** | Category-specific automotive news articles linked to relevant listings |

| ****360° View Support**** | Listings can embed interactive 360-degree vehicle views |

| ****Shorts / Video Reels**** | Short vertical videos attached to listings for better discovery |

| ****Social Sharing**** | One-click share listings to WhatsApp, Facebook, Twitter etc. |

🏠 1.2 Dealer Portal

Designed for dealerships of all types — new vehicles, resale, accessories, spare parts, and rentals.

| Feature | Description |

|---|---|

| ****Separate Dealer Login**** | Dealers have their own protected login (`/dealer/login`) with credential-based access |

| ****Multi-Type Dealer Support**** | Supports 5 dealer types: New Car/Bike, Resale, Accessories, Spare Parts, Rental |

| ****Dealer Inventory Management**** | Separate modules: `dealer-accessories`, `dealer-parts`, `dealer-rental`, `dealer-used` |

| ****Inquiry CRM**** | All test drive and enquiry leads from platform users automatically routed to matching dealers by pincode and brand |

| ****Showroom Assignment**** | Admin assigns incoming user inquiries to the correct showroom based on area pin codes |

| ****Test Drive Management**** | Accept or schedule test drives (At Showroom or At Home) |

| ****Variant & Pricing Config**** | Set up multiple variants per vehicle with individual RTO, insurance, and ex-showroom pricing |

| ****Media Uploads**** | Photos, videos, shorts, and exterior/interior gallery management via Cloudinary |

| ****Listing Offers**** | Add time-limited offers and discounts to any listing |

| ****Standout Feature Cards**** | Highlight unique selling points of a listing with rich image-based cards |

| ****Expert Review Panel**** | Add professional expert reviews for each vehicle listing |

🔑 1.3 Reseller Portal

A unique zone-locked reselling program for verified individual resellers.

| Feature | Description |

|---|---|

| ****Reseller Registration**** | Resellers apply with Business Name, PAN, Mobile, and vehicle number for verification |

| ****Zone Exclusivity**** | A maximum of ****5 resellers per pincode area****, creating scarcity and premium value |

| ****PAN Verification**** | Owner PAN encrypted and stored securely using AES-256 encryption |

| ****Listing Limits**** | Each reseller has a configurable max listing count (default: 50) |

| ****Status Tracking**** | Reseller applications are tracked (Pending → Approved → Suspended/Rejected) |

🛡️ 1.4 Admin Panel

Full command-and-control panel for platform governance, accessible only to verified admin email.

Feature	Description
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Dashboard Overview	Live metrics: Total Users, Total Listings, Pending Approvals, Enquiries, Test Drives
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Listing Approvals	Review, approve, reject or delete any user, dealer, or scraped listing
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User Management	View all users, change roles (user/admin/dealer/brand), ban/unban, delete accounts
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Dealer Account Creation	Admin creates dealer logins and configures their profile, type, and access
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Web Scraper Tool	Import vehicle listings from external sources like CarDekho using the built-in Cheerio/Puppeteer scraper
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Enquiry Management	View and manage all customer enquiries and assign them to the right showroom
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Test Drive Management	View all pending test drive requests and their statuses
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Category Management	Create, edit, or remove accessory/spare parts categories
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Contact Messages	View all messages submitted via the Contact form
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News Management	Create news articles and link them to specific vehicle listings
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Service Station Creation	Register new service stations with location, services, hours, and pricing
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Admin Action Logs	Every admin action (approve, ban, update, import, delete) is permanently logged for auditability
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Chat Analytics Dashboard	Track AI chatbot performance: total conversations, intent accuracy, failed intents, satisfaction scores
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1.5 Service Station Portal

A dedicated module for automotive service centres to participate in the platform.

| Feature | Description |

|---|---|

| ****Station Profiles**** | Full station profiles with name, address, GPS coordinates, working hours, phone and email |

| ****Vehicle Type Config**** | Configure whether station serves Cars, Bikes, or Both |

| ****Service Catalogue**** | Map specific services (oil change, tyre, brake, etc.) with optional pricing |

| ****Pricing Models**** | Supports Fixed pricing, Range pricing, or Inspection-based pricing |

| ****Booking Types**** | Supports At-Centre and At-Home service bookings |

| ****Review & Rating System**** | Customer reviews with star ratings, admin-moderated, tied to verified user accounts |

| ****Inquiry Management**** | Receive service booking inquiries with customer name, phone, pincode, and selected services |

2. UNIQUE SELLING POINTS

USP 1 — All-in-One Automotive Ecosystem

Car & Bike is the ****only Indian automotive platform**** covering new vehicles, used vehicles, rentals, accessories, spare parts, service stations, and reseller programs all in one product.

📍 USP 2 — Pincode-First Hyper-Local Experience

Unlike competitors, every experience — listings, dealers, service stations, offers — is personalised to the user's ****exact pincode****. Dealers are matched to buyers by geographic proximity automatically.

🗣️ USP 3 — Built-In AI Car Assistant

A live conversational AI chatbot (powered by Twilio) is embedded on the platform. It handles 24/7 queries, assists with search, and captures intent data to help the platform learn and improve.

📺 USP 4 — Video-First Listings

Support for ****Shorts (vertical videos)****, YouTube embeds, full video galleries, and 360-degree views — making Car & Bike the most media-rich vehicle listing platform available.

🛡️ USP 5 — Verified Ecosystem

Users must verify with Aadhaar or PAN. Dealers go through admin approval. Resellers are PAN-verified. All listings require admin sign-off. This creates a ****trusted, fraud-resistant marketplace****.

🔍 USP 6 — Web Scraper for Inventory Seeding

Admin can import listings from competitors like CarDekho using a built-in scraper — allowing the platform to ****bootstrap inventory instantly**** without waiting for organic listings.

💰 USP 7 — Multi-Revenue Stream Architecture

The platform generates income from ****multiple streams simultaneously****: listing fees, rental commissions, dealer subscriptions, reseller programme fees, service booking commissions, and featured listing placement.

3. WHY CHOOSE CAR & BIKE OVER CARDEKHO, BIKEWALE & OLX AUTOS?

Feature	**Car & Bike**	CarDekho	BikeWale	OLX Autos	Cars24
New Cars & Bikes	✓	✓	✓ (bikes only)	✗	✗
Used Cars & Bikes	✓	✓	✗	✓	✓
Vehicle Rentals	✓	✗	✗	✗	✗
Accessories Shop	✓	✗	✗	✗	✗
Spare Parts	✓	✗	✗	✗	✗
Service Station Booking	✓	✗	✗	✗	✗
AI Chatbot (24/7)	✓	✗	✗	✗	✗
Pincode-Based Localisation	✓	Partial	✗	✗	Partial
Shorts / Reels on Listings	✓	✗	✗	✗	✗
360° Vehicle View	✓	✗	✗	✗	✗
Verified Reseller Programme	✓	✗	✗	✗	✗
Admin Web Scraper	✓	N/A	N/A	N/A	N/A
Full On-Road Price Breakdown	✓	✓	Partial	✗	✓
Separate Dealer Portal	✓	Partial	Partial	✗	✗
AES-256 Document Encryption	✓	✗	✗	✗	✗
Admin Action Audit Logs	✓	✗	✗	✗	✗

Key Differentiators Explained

****1. CarDekho is only a lead generator. Car & Bike is a complete marketplace.****

CarDekho primarily redirects users to external dealer websites and collects a lead fee. Car & Bike facilitates the entire journey — discovery, inquiry, test drive booking, purchase, and post-sale service — within one platform.

****2. OLX Autos has zero trust layers. Car & Bike has a multi-step verification system.****

OLX Autos is an open classifieds platform. Any user can post any vehicle with no checks. Car & Bike requires document verification for sellers, admin approval for listings, and PAN verification for resellers.

****3. No competitor has a rental + accessories + spare parts stack.****

After buying a vehicle from our platform, the same user can rent a replacement during servicing, buy accessories, and order spare parts. This is unprecedented in India.

****4. Service station integration is a completely untapped vertical.****

Indian automotive marketplaces have entirely ignored post-sale service. Car & Bike integrates local service stations with bookable slots, reviews, and service catalogues — creating recurring revenue and loyalty.

****5. Pincode intelligence beats city-level filtering.****

Most platforms filter by city (e.g., "Mumbai"). Car & Bike filters by 6-digit pincode, connecting buyers with dealers within their literal neighbourhood, dramatically increasing lead quality.

4. TECHNOLOGY STACK

| Layer | Technology |

|---|---|

| **Framework** | Next.js 14 (App Router, SSR, Streaming with Suspense) |

| **Language** | TypeScript |

| **Databases** | MySQL (Prisma ORM) + MongoDB (Mongoose) |

| **Authentication** | NextAuth v4 — Google OAuth 2.0 + Admin credentials |

| **File Uploads / CDN** | Cloudinary (images, videos, shorts) |

| **Styling** | TailwindCSS with dark theme + Framer Motion animations |

| **AI Chatbot** | Twilio AI Conversations SDK |

| **Web Scraping** | Cheerio + Puppeteer |

| **Forms / Validation** | React Hook Form + Zod |

| **Encryption** | AES-256 for sensitive documents (Aadhaar, PAN) |

| **Email / SMS** | Twilio Communications |

| **Social Sharing** | next-share library |

| **SEO** | Dynamic metadata generation, Open Graph, Twitter Cards |

| **Testing** | Vitest (unit), Playwright (E2E) |

| **Hosting** | Vercel-ready (also supports AWS, DigitalOcean, cPanel) |

5. DATABASE ARCHITECTURE

The platform is backed by a **35+ model relational database** covering:

- Users, Roles, Verification Documents
- Listings (Vehicles, Accessories, Spare Parts, Rentals)
- Listing Media, Variants, Specifications, Offers, Comparisons, News
- Inquiries, Test Drives, Showroom Assignments
- Reseller Profiles, Brand Partners
- Service Stations, Services, Bookings, Reviews
- Notifications, Conversations, Chat Analytics
- Admin Logs, Pincodes, Categories, Contact Messages

All structured with performance indexes on key query paths (status, city, price, brand, pincode).

6. SECURITY & COMPLIANCE

| Security Feature | Implementation |

|---|---|

| Google OAuth + NextAuth Session | Industry standard authentication |

| AES-256 Document Encryption | Aadhaar and PAN data encrypted before storage |

| Admin Route Protection | Server-side session check on all admin pages |

| Dealer Route Protection | Credential-based auth for dealer portal |

| Input Validation | Zod schemas on all API routes |

| Admin Action Logs | Immutable audit log for every admin operation |

| Environment Secrets | All sensitive config in `.env.local`, never committed |

7. DEPLOYMENT & SCALABILITY

- Fully ready for **Vercel** deployment (recommended for Next.js)
- Supports **cPanel MySQL hosting** (currently configured)
- MongoDB Atlas for scalable NoSQL operations
- Cloudinary for CDN-backed media delivery at global scale
- Memory-optimised build with `--max-old-space-size=4096`
- Node Cache for server-side performance optimisation

8. INVESTMENT POTENTIAL

Revenue Streams

1. **Dealer Subscriptions** — Monthly fee for dealerships to list vehicles
2. **Featured Listings** — Premium placement for highlighted listings
3. **Lead Fees** — Per-inquiry monetisation when leads are sent to dealers
4. **Rental Commission** — Percentage of each rental booking
5. **Reseller Programme Fees** — Onboarding fee for zone-locked resellers
6. **Service Station Commissions** — Percentage of service bookings
7. **Accessory & Spare Parts Sales** — Transactional eCommerce margin

Target Market

- 300+ million two-wheeler owners in India
- 35+ million car owners growing annually
- 15,000+ registered car dealerships nationwide
- Massive used vehicle ecosystem (~₹1.5 lakh crore market)

9. CONCLUSION

****Car & Bike**** is not just another automotive listing website. It is a ****complete digital ecosystem**** for the ₹1.5+ lakh crore Indian automotive market — combining buying, selling, renting, servicing, and accessorising in a single, beautifully designed, technologically advanced platform.

Where CarDekho stops at a lead form, Car & Bike closes the entire deal.

Where OLX Autos has zero trust, Car & Bike verifies every participant.

Where no competitor offers rentals + service + spare parts — ****Car & Bike offers all of it.****

We invite you to partner with us and capture a dominant position in one of India's fastest-growing digital commerce categories.

****For demos, pricing, and partnership discussions:****

✉ admin@carandbike.com

🌐 www.carandbike.com

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